For this assignment I chose the Animal Shelter Dashboard I did from my CS340 course. This project was made using Python and MongoDB to store animal shelter data and then display it in a dashboard. The dashboard lets someone view all animal records and filter them based on the rescue types as well as see all the information in a graph, table, and map. I chose to do this one because it already worked with a database and there were a lot of ways it could be improved on.

I chose to do this because it shows my skills working with databases and that I know how to use them with a program to store and retrieve data. One of the things I changed was I added pagination so the program doesn't have to get all the animal records from MongoDB every time the table is displayed and instead it just gets the records needed for the current page its on. I also added a compound index for the animal type, breed, and age so MongoDB has a better way to search through all the data. I added a search bar that lets people search by the animals name, breed, or ID and added input handling so there's no issues with special characters.

Another thing I did was use MongoDB aggregation for the breed graph. The graph was originally based only on the records that were displayed in the table so it would only show information for the 10 animals on that page so I changed it so MongoDB groups and counts all the breeds from all of the data that match the filters or the search and then displays the top 10 breeds so everything stays readable even when all the data is being used. I also improved on the security of the database by removing the username and password that were hardcoded and used environment variables instead.

I think I met my outcome goal that I had originally planned. Outcomes 3 and 4 are still good because I improved how the program works with the database and I used different database techniques to make searching all the data a lot easier. I was also able to use MongoDB indexing, pagination, searching, and aggregation instead of just having all of the information loaded all at once. The changes I did with the database login also gave me more experience on how I think about and handle security and how sensitive information should be dealt with. I think I met all outcomes and those outcomes still cover the database skills I showed.

Working on this helped me understand more about how a database can be used to do more of the work instead of only having them process everything after you give it all the data. Before making the changes I never really thought about how loading so much data at once can be a problem especially as it just keeps getting larger Adding pagination and aggregation showed me how the program can only get the information it actually needs and how MongoDB can group and count all the data before it sends it all back.

A challenge I had was that the original zip I had was missing the CRUD Python module that it needed to connect the dashboard to MongoDB but I did end up finding the original code online and then added the new database methods needed for the enhancement. I was also having issues adding database methods because I would make a small mistake and would have the wrong indentation somewhere and it would end up outside of the CRUD class so it took a while to find where I went wrong. Another issue I was having was because I had everything open like visual

studio and Jupyter, all the changes I made would sometimes be reverted or overwritten and I had to figure out how to keep the saved version I wanted updated between both of the programs. I did a lot of testing on this like the rescue filters, searches, pagination, the database index, and aggregation to make sure everything worked correctly after each change I did. Working through all of this gave me more experience debugging a project and also showed me why it's important to keep all the files together.